Chapter 4: Regression

School of Mathematics and Computational Sciences
University of Prince Edward Island

Instructor: Ali Raisolsadat

Part I: Exploring Data

Why Regression Matters

Chapter 3 introduced scatterplots and correlation as tools for describing the direction and strength of a linear association between two quantitative variables.

Regression goes one step further. It gives an equation for a fitted line that can be used to describe how a response variable tends to change as an explanatory variable changes.

In first-year statistics, regression is important because it connects visual patterns, formulas, prediction and interpretation .

From Correlation to Regression

Correlation and regression are related but they answer different questions.

Idea	Correlation	Regression
Main purpose	Describes the direction and strength of a linear association	Describes the fitted relationship and predicts the response
Roles of variables	The two variables are treated symmetrically	One variable is explanatory and one is the response
Units	Unitless	The slope has response units per explanatory-variable unit
Typical question	How strong is the linear association	What response is predicted for a given value of x

The Roles of the Variables

A regression analysis begins by assigning roles to the variables.

- The explanatory variable x is used to explain or predict
- The response variable y is the outcome being described or predicted

These roles come from the scientific question rather than from the order of the columns in a dataset.

A fitted line describes an average pattern. Two individuals with the same value of x can still have different observed responses.

Regression is a Model

A fitted regression line summarizes a linear pattern in the observed data.

The fitted line does not guarantee that every point lies on the line.
It does not establish that changes in the explanatory variable cause changes in the response variable.

A useful interpretation should therefore distinguish clearly between

- description of the observed relationship
- prediction from the fitted line
- causal conclusions

The Anatomy of a Straight Line

A straight line relating an explanatory variable x to a response variable y has the form

$$y = a + bx$$

The slope b describes the change in the predicted response for a one-unit increase in x.

The intercept a is the value of the line when $x = 0$.

From an Algebraic Line to a Regression Line

A regression line uses the same algebraic form but the predicted response is written as $\hat{y}$.

$$\hat{y} = a + bx$$

The notation matters.

- y is an observed response taken from the data
- $\hat{y}$ is the response predicted by the fitted line

This distinction becomes very important when residuals are introduced.

How the Slope and Intercept Are Read

The slope and intercept do different jobs.

- The intercept locates the line vertically
- The slope determines how quickly the fitted response rises or falls as x changes

The units of the slope are the units of the response variable divided by the units of the explanatory variable.

The units of the intercept are the units of the response variable.

Reading a Regression Equation

Question

A greenhouse study uses the fitted equation

$$\widehat{\text{height}} = 4.5 + 1.8(\text{hours of light})$$

to predict final plant height in centimetres from daily light exposure in hours.
What does the slope mean? What height does the model predict for a plant receiving 6 hours of light per day?

Reading a Regression Equation

The slope is 1.8 centimetres per hour of light.

According to the fitted linear model, a one-hour increase in daily light exposure is associated with an increase of 1.8 centimetres in predicted final height.

The intercept is 4.5 centimetres. It is the predicted height when the explanatory variable equals zero. Whether that interpretation is useful depends on the scientific setting.

Additional Example: Solution

For 6 hours of light, the predicted height is

$$\widehat{\text{height}} = 4.5 + 1.8(6)$$

$$\widehat{\text{height}} = 15.3 \text{ cm}$$

The fitted model predicts a final height of 15.3 cm for a plant receiving 6 hours of light per day.

Example 4.1: Predicting Fat Gain

Question

A study followed 16 young adults who overate for eight weeks.
The explanatory variable was the change in nonexercise activity (NEA) measured in Calories.
The response variable was fat gain measured in kilograms.
How can the fitted regression line be used to predict fat gain for a person whose NEA increases by 400 Calories?

Example 4.1: The Observed Data

Volunteer	NEA Change (Calories)	Fat Gain (kg)
1	-94	4.2
2	-57	3.0
3	-29	3.7
4	135	2.7
5	143	3.2
6	151	3.6
7	245	2.4
8	355	1.3

Volunteer	NEA Change (Calories)	Fat Gain (kg)
9	392	3.8
10	473	1.7
11	486	1.6
12	535	2.2
13	571	1.0
14	580	0.4
15	620	2.3
16	690	1.1

Example 4.1: Explanation

The fitted least-squares regression line for these data is

$$\widehat{\text{fat gain}} = 3.505 - 0.00344(\text{NEA change})$$

A prediction at an NEA change of 400 Calories is obtained by substituting $x = 400$ into the fitted equation.

The value 400 lies inside the observed range of NEA changes, so this prediction is an interpolation rather than an extrapolation.

Example 4.1: Solution

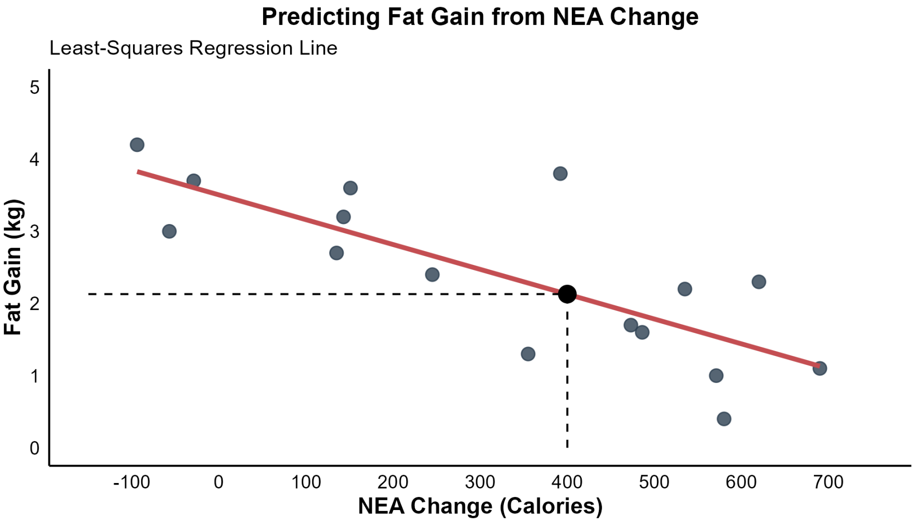
The predicted fat gain is

$$\widehat{\text{fat gain}} = 3.505 - 0.00344(400)$$

$$\widehat{\text{fat gain}} \approx 2.13 \text{ kg}$$

The value 2.13 kg is the response predicted by the fitted line for a person whose NEA increases by 400 Calories.

Example 4.1: Scatterplot and Fitted Line



Reading the Context

The slope is -0.00344 kg per Calorie.

Within the range of NEA values represented by the data, the fitted model predicts that fat gain decreases by about 0.00344 kg for each additional Calorie of NEA change.

A 100-Calorie increase in NEA corresponds to a predicted decrease of about 0.344 kg in fat gain according to the fitted line.

The Meaning of the Intercept

The intercept is 3.505 kg. It represents the predicted fat gain when NEA change is zero.

$$\widehat{\text{fat gain}} = 3.505 \text{ kg when NEA change} = 0$$

In this example, the value $x = 0$ lies inside the observed range, so the intercept has a meaningful interpretation.

In other applications, the intercept may be mathematically necessary while having little practical meaning.

A Reminder About Units

The numerical size of a slope depends on the units.

The slope -0.00344 does not imply that the relationship is unimportant simply because the number is close to zero.

If fat gain were measured in grams instead of kilograms, the slope would be -3.44 grams per Calorie. The fitted relationship would be unchanged.

Least-Squares Regression

A line drawn by eye is subjective. Different people could draw slightly different lines through the same scatterplot.

Least-squares regression provides a precise mathematical rule for choosing one line.

For an observation (x_i, y_i), the fitted line gives a predicted response $\hat{y}_i$. The residual is the signed vertical difference between the observed response and the predicted response.

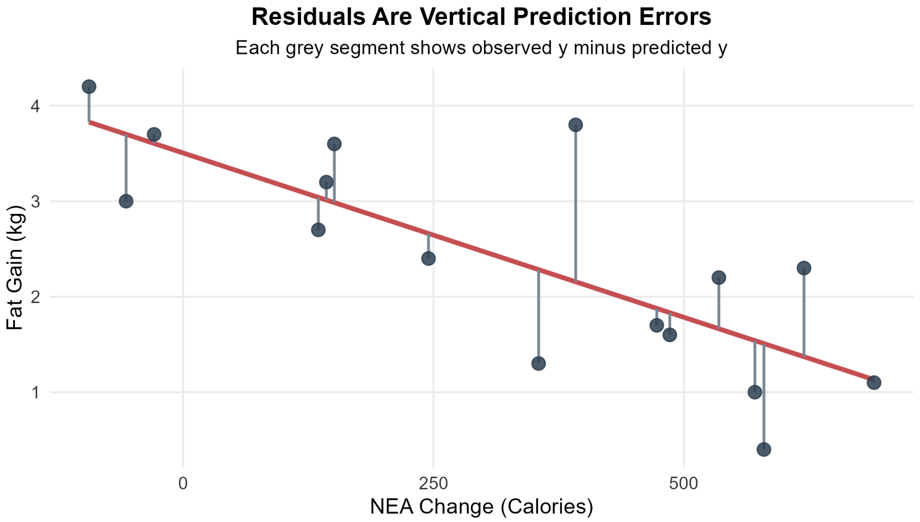
$$e_i = y_i - \hat{y}_i$$

How Residuals Are Interpreted

A residual compares what was observed with what the fitted line predicted.

- A positive residual means the observed point lies above the fitted line
- A negative residual means the observed point lies below the fitted line
- A residual close to zero means the fitted line predicted that observation well

Residual Plot Illustration



Residuals from the NEA Study

Question

Volunteer 8 had an NEA change of 355 Calories and an observed fat gain of 1.3 kg. What fat gain does the fitted line predict for this volunteer and what is the residual?

Residuals from the NEA Study

The fitted value is

$$\hat{y} = 3.505 - 0.00344(355)$$

$$\hat{y} \approx 2.28 \text{ kg}$$

The residual is

$$e = y - \hat{y} = 1.3 - 2.28$$

$$e \approx -0.98 \text{ kg}$$

The residual is negative. So this point lies below the fitted line.

Why Are the Residuals Squared

The least-squares line is the line that minimizes the sum of squared residuals.

$$\text{Sum of squared residuals} = \sum e_i^2$$

Squaring serves two purposes.

- Positive and negative residuals do not cancel out
- Large prediction errors are penalized more strongly than small ones

Notation and Formula Reference

Least-Squares Regression Formulas

$$b = r\frac{s_y}{s_x}$$

$$a = \bar{y} - b\bar{x}$$

$$\hat{y} = a + bx$$

$$e = y - \hat{y}$$

These formulas connect the regression coefficients to the summary statistics from Chapter 3.

What the Slope Formula Tells Us

The slope formula

$$b = r\frac{s_y}{s_x}$$

shows that the slope depends on three ingredients.

- The sign and strength of the linear association through r
- The spread of the response variable through s_y
- The spread of the explanatory variable through s_x

A positive correlation gives a positive slope. A negative correlation gives a negative slope.

Example 4.2: Calculating the Line from Summary Statistics

Question

For the NEA and fat-gain data, the summary statistics are

$$\bar{x} = 324.75$$

$$\bar{y} = 2.3875$$

$$s_x = 261.46$$

$$s_y = 1.1207$$

$$r = -0.8021$$

Use these summary statistics to calculate the least-squares regression line.

Example 4.2: Calculate the Slope

Using the slope formula,

$$b = r\frac{s_y}{s_x}$$

$$b = (-0.8021)\left(\frac{1.1207}{261.46}\right)$$

$$b \approx -0.00344$$

The fitted line has a negative slope, which agrees with the negative association seen in the scatterplot.

Example 4.2: Calculate the Intercept

Using the intercept formula,

$$a = \bar{y} - b\bar{x}$$

$$a = 2.3875 - (-0.00344)(324.75)$$

$$a \approx 3.505$$

The least-squares regression line is therefore

$$\widehat{\text{fat gain}} = 3.505 - 0.00344(\text{NEA change})$$

Calculating the Least-Squares Line in R

R can reproduce the hand calculations and can also fit the model directly.

The same data are used in each method so that the connection between the formulas and the software remains clear.

R Code: Entering the Data

```
# NEA change in Calories
x <- c(
 -94, -57, -29, 135, 143, 151, 245, 355,
  392, 473, 486, 535, 571, 580, 620, 690
)

# Fat gain in kilograms
y <- c(
  4.2, 3.0, 3.7, 2.7, 3.2, 3.6, 2.4, 1.3,
  3.8, 1.7, 1.6, 2.2, 1.0, 0.4, 2.3, 1.1
)
```

R Code: Step-by-Step Calculation

```
# Summary statistics
x_bar <- mean(x)
y_bar <- mean(y)
s_x <- sd(x)
s_y <- sd(y)
r <- cor(x, y)

# Least-squares coefficients
b <- r * (s_y / s_x)
a <- y_bar - b * x_bar
```

R Code: A Custom Function

```
calc_regression <- function(x_var, y_var) {
  b_slope <- cor(x_var, y_var) * (sd(y_var) / sd(x_var))
  a_intercept <- mean(y_var) - b_slope * mean(x_var)

  data.frame(
    Mean_X = mean(x_var),
    Mean_Y = mean(y_var),
    Correlation = cor(x_var, y_var),
    Slope = b_slope,
    Intercept = a_intercept
  )
}

calc_regression(x_var = x, y_var = y)
```

R Code: Fitting the Model with `lm()`

```
model <- lm(y ~ x)

# Display the fitted coefficients
coef(model)
```

The formula $y \sim x$ means that the response variable `y` is modelled as a function of the explanatory variable `x`.

R Code: A Prediction from the Fitted Model

```
# Predicted fat gain for an NEA change of 400 Calories
predict(model, newdata = data.frame(x = 400))
```

This R output agrees with the earlier prediction of about 2.13 kg.

R Code: Fitted Values and Residuals

```
# Fitted responses and residuals from the linear model
fitted_values <- fitted(model)
residual_values <- residuals(model)

# The first six observations are displayed together
head(
  data.frame(
    NEA = x,
    Observed_Fat_Gain = y,
    Predicted_Fat_Gain = fitted_values,
    Residual = residual_values
  )
)
```

R Code: Model Summary

```
# A standard summary of the fitted linear model
summary(model)
```

At this stage, the fitted coefficients and R^2 are the most relevant parts of the output.

Facts About Least-Squares Regression

Several mathematical properties help explain how a fitted line behaves.

- The roles of x and y are not interchangeable
- The sign of the slope always matches the sign of the correlation
- The least-squares line always passes through $(\bar{x}, \bar{y})$
- The coefficient of determination R^2 summarizes how well the line accounts for variation in the response

The Roles of x and y

The least-squares line for predicting y from x is generally different from the least-squares line for predicting x from y.

This difference occurs because the regression line minimizes vertical prediction errors in the response variable.

Regression is therefore directional in a way that correlation is not.

Slope and Correlation

The slope and correlation are connected by the formula

$$b = r\frac{s_y}{s_x}$$

The sign of the slope always matches the sign of r.

- If $r > 0$, then $b > 0$
- If $r < 0$, then $b < 0$
- If $r = 0$, then $b = 0$

The Point of Averages

The least-squares regression line always passes through the point of averages.

$$(\bar{x}, \bar{y})$$

This fact is built into the intercept formula

$$a = \bar{y} - b\bar{x}$$

It provides a useful algebraic check when the fitted line is calculated by hand.

The Coefficient of Determination

The coefficient of determination is written as R^2.

For simple linear regression with an intercept,

$$R^2 = r^2$$

The value R^2 is the proportion of the observed variation in the response variable that is accounted for by the fitted linear relationship with the explanatory variable.

A Second View of R^2

A useful interpretation of R^2 compares total variation with variation left unexplained by the model.

$$R^2 = 1 - \frac{\text{SSE}}{\text{SST}}$$

Here, SSE is the sum of squared residuals and SST is the total sum of squares.

A larger R^2 means the fitted line leaves less variation unexplained.

Example 4.3: Interpreting R^2

Question

Suppose a fitted linear model has

$$R^2 = 0.75$$

How should this value be interpreted?

Example 4.3: Solution

An R^2 value of 0.75 means that 75% of the observed variation in the response variable is accounted for by the fitted linear relationship with the explanatory variable.

The remaining 25% is not accounted for by the fitted line and remains in the residual variation.

This interpretation describes the quality of fit. It does not imply causation.

Outliers and Influential Observations

A single unusual observation can affect correlation and least-squares regression in important ways.

Three related ideas should be distinguished.

- An outlier has an unusually large residual
- A high-leverage point has an unusual value of x
- An influential point changes the fitted line noticeably when it is included or removed

Examples 4.4 and 4.5: The Empathy Study

Question

An empathy study measures brain activity and empathy score for 16 women. Subject 16 has an unusually large empathy score.
How does this point affect the correlation and the least-squares regression line?

Examples 4.4 and 4.5: Interpretation

Subject 16 changes the correlation substantially.

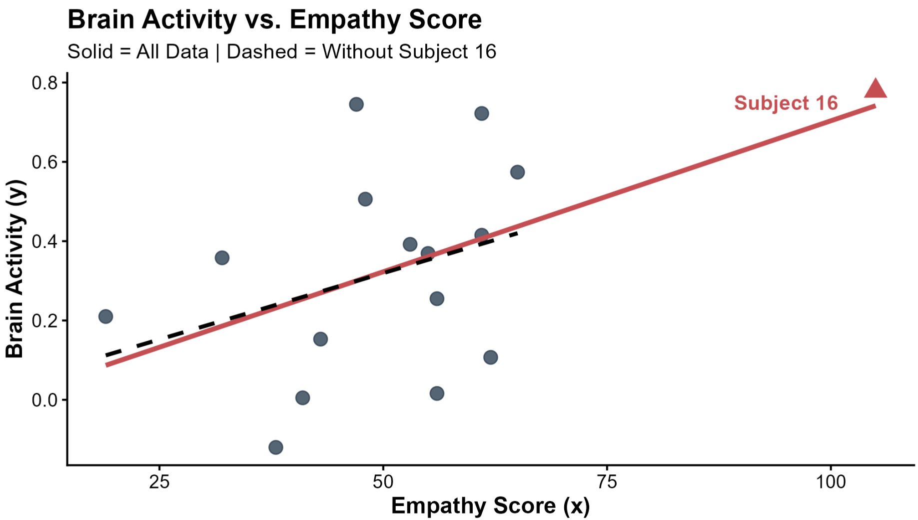
$$r = 0.515 \text{ with Subject 16}$$

$$r = 0.331 \text{ without Subject 16}$$

The fitted regression line also changes, although less dramatically than the correlation.

This example shows that a point can be influential for both correlation and regression, with different levels of impact.

Empathy Study Plot



A Hypothetical High-Leverage Point

A point with an extreme x value has leverage because it can pull the fitted line toward itself.

If a high-leverage point also has a large residual, it can become highly influential.

This is why unusual observations should be examined carefully before a fitted line is interpreted.

Why Use a Logarithm Transformation

In biology and other sciences, data often span several orders of magnitude.

A logarithm transformation can help when

- the raw data are strongly skewed
- a few very large observations dominate the scale
- a multiplicative relationship becomes more nearly linear after transformation

A Simple Base-10 Log Illustration

A base-10 logarithm compresses large scales.

$$\log_{10}(10) = 1$$

$$\log_{10}(100) = 2$$

$$\log_{10}(1000) = 3$$

$$\log_{10}(10000) = 4$$

Values that are very far apart on the original scale become much closer on the log scale.

Example 4.6: Mammal Brains and Body Weight

Question

How does an animal's brain size relate to its body size across many mammal species?
Why might a log-log transformation give a clearer regression picture than the raw variables?

Example 4.6: Explanation

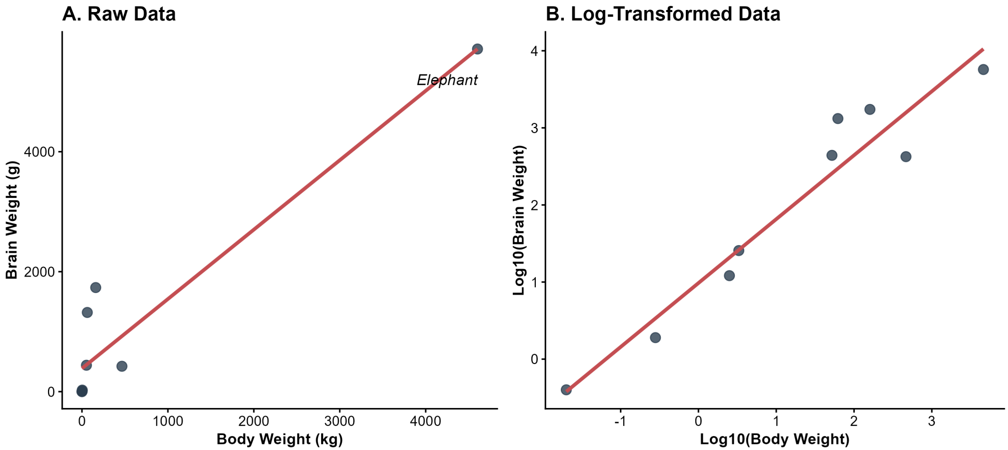
On the raw scale, a few very large mammals dominate the axes and compress the smaller species near the origin.

After taking base-10 logarithms of both variables, the observations are spread more evenly and the relationship becomes more nearly linear.

The fitted log-log model is

$$\log_{10}(\text{brain weight}) = 1.01 + 0.72 \log_{10}(\text{body weight})$$

Mammal Example Plot



Back to the Original Scale

The log-log model can be rewritten as a power relationship.

$$y = 10^{1.01} x^{0.72}$$

This form shows that a multiplicative relationship on the original scale becomes linear after taking logarithms.

Cautions About Correlation and Regression

A fitted line should always be interpreted in light of the data and the scientific question.

- Correlation and simple linear regression describe linear patterns
- Unusual observations can change the correlation and the fitted line
- Predictions outside the observed range require caution
- A lurking variable can help explain an observed association
- Association by itself does not imply causation

Linear Relationships

A scatterplot should be examined before a simple linear model is interpreted.

Curvature, separate clusters or changing spread can all indicate that a straight line is an inadequate summary.

R or other statistical software can always produce a correlation coefficient and a fitted line, but those outputs are only useful when the relationship is reasonably linear.

Interpolation and Extrapolation

A prediction within the observed range of x values is called interpolation.

A prediction outside the observed range is called extrapolation.

Algebra allows the fitted line to be extended indefinitely, but the data do not show whether the same relationship continues beyond the observed range.

Additional Example: Interpolation versus Extrapolation

Question

A fertilizer study observes daily doses from 0 to 5 millilitres and fits a linear regression model over that range.
Which prediction is better supported by the observed data: a prediction at 3 millilitres or a prediction at 100 millilitres?

Additional Example: Solution

The prediction at 3 millilitres is better supported because it is an interpolation.

The prediction at 100 millilitres is an extrapolation far beyond the observed data.

The fitted line contains no information about whether the biological relationship remains linear from 5 to 100 millilitres.

Lurking Variables

A lurking variable is a variable that is not included as an explanatory or response variable in the analysis but may help explain the observed association.

For example, an observational study might find that coffee consumption is positively associated with risk of heart attack. In this case, smoking could be a lurking variable if it is related to both coffee consumption and heart attack risk.

Association Does Not Imply Causation

Regression and correlation describe associations. They do not by themselves establish cause and effect.

A causal conclusion requires more than a strong correlation, a steep slope or a large R^2. The study design matters.

Evidence for Causation

A well designed randomized experiment can provide stronger evidence for a causal effect because random assignment helps balance lurking variables across treatment groups.

Observational studies generally require more caution because unmeasured confounding can remain even after statistical adjustment.

A Consistent Regression Workflow

A useful first-year workflow is to interpret regression in the same order every time.

1. Identify the explanatory variable and the response variable
2. Examine the scatterplot for linearity and unusual observations
3. Write and interpret the fitted line
4. Calculate or interpret a prediction when appropriate
5. Use residuals and R^2 to assess the fit
6. Consider limitations such as extrapolation, lurking variables and causation